

Basic Logical Connectives

Topics of this Lesson

- Negation (Not)
- Conjunction (And)
- Disjunction (Or)
- Implication or if-then (Conditional)
- If and Only if (Equivalence or biconditional)

Negation

- It means the opposite of the original statement. If p is a statement, then the negation of p is denoted by $\sim p$ and read as 'it is not the case that p , or simply Not P .' So, if p is true then $\sim p$ is false and vice versa.
- Example:
 - If the statement P is Paris is in France, then $\sim P$ is 'Paris is not in France'.
 - R : Today David is present in the class
 - $R = \{T, F\}$
 - $\sim R$: Today David is not present in the class

Truth Table of Negation

p	$\sim p$
T	F
F	T

Conjunction

- It means AND operation of two statements. If p , q are two statements, then " p and q " is a compound statement, denoted by $p \wedge q$ and referred as the conjunction of p and q . The conjunction of p and q is true only when both p and q are true. Otherwise, it is false.
- Example:
 - P : John went shopping yesterday.
 - R : He bought a shirt.
 - Q : The color of the shirt is blue.
 - $P \wedge R$: John went for shopping yesterday and he bought a shirt.

Truth Table of Conjunction

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

P: John went for shopping yesterday	R: He bought a shirt	$P \wedge R$
F	F	F
F	T	F
T	F	F
T	T	T

Disjunction

- It means OR operation of two statements. If p , q are two statements, then " p or q " is a compound statement, denoted by $p \vee q$ and referred to as the disjunction of p and q . The disjunction of p and q is true whenever at least one of the two statements is true, and it is false only when both p and q are false.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Implication / if-then ($\longrightarrow$)

- An implication $p \longrightarrow q$ is the proposition "if p , then q ." It is false if p is true and q is false. The rest of the cases are true.

p	q	$p \longrightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

If you study (T)	You will do well (T)	True
If you do not study (F)	You will not do well (F)	True
You did not study well (F)	You did well in the exam (T)	True
You studied well (T)	You did not do well in the exam (F)	False

Bi-conditional / if and only if ($\leftrightarrow$)

- $p \leftrightarrow q$ is bi-conditional logical connective which is true when p and q are same, i.e., both are false or both are true.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T